LECTURE SERIES · DATABASE SYSTEMS · WEEK 5

Triggers, Backup & Restore

PostgreSQL in Practice

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Stored Procedures · Functions · Triggers · Views · Materialized Views

T

AUTOMATIC ACTIONS ON DATA CHANGES

What is a Trigger?

A trigger runs a function automatically when a table is modified. Define it in two steps: a trigger function, then **CREATE TRIGGER**.

Event

What fires it.

```
INSERT | UPDATE
DELETE | TRUNCATE
```

Timing

When it runs.

```
BEFORE | AFTER
INSTEAD OF (views)
```

Level

How often.

```
FOR EACH ROW
FOR EACH STATEMENT
```

BEFORE vs AFTER

BEFORE can change NEW, or RETURN NULL to skip the row. **AFTER** sees the final result, ideal for audit logs.

SPECIAL TRIGGER VARIABLES

NEW and OLD

Operation	NEW	OLD
INSERT	The new row	Not available
UPDATE	Row after the change	Row before the change
DELETE	Not available	The row being deleted

NEW

Row after the change.

```
NEW.salary
```

OLD

Row before the change.

```
OLD.salary
```

TG_OP

Which event fired.

```
IF TG_OP = 'DELETE'
```

CONTROLLING TRIGGERS

Manage Triggers

Inspect, pause or remove triggers without touching the data.

List

Show user-defined triggers.

```
SELECT tgname FROM pg_trigger
WHERE NOT tgisinternal;
```

Disable / Enable

Pause one temporarily.

```
ALTER TABLE employee
DISABLE TRIGGER trg_salary_audit;
```

Drop

Remove it for good.

```
DROP TRIGGER trg_salary_audit
ON employee;
```

Cancel a row

In a BEFORE trigger.

```
RETURN NULL; -- skip row
RETURN NEW; -- continue
```

PG_DUMP IN THE CLI

Database Backup

pg_dump exports one database while it stays online. Pick the format that fits the restore.

Plain SQL

Readable script.

```
pg_dump -U postgres -d mydb \
  -f mydb.sql
```

Custom format

Compressed, flexible.

```
pg_dump -U postgres -Fc \
  -d mydb -f mydb.dump
```

Directory, parallel

Fast for large DBs.

```
pg_dump -U postgres -Fd -j 4 \
  -d mydb -f mydb_dir
```

All databases

Plus roles and globals.

```
pg_dumpall -U postgres \
  -f all.sql
```

PSQL AND PG_RESTORE IN THE CLI

Database Restore

Plain SQL goes through **psql**. Custom and directory formats need **pg_restore**.

Plain SQL

Run the script.

```
createdb -U postgres mydb_new
psql -U postgres -d mydb_new \
  -f mydb.sql
```

Custom format

Restore the .dump file.

```
pg_restore -U postgres \
  -d mydb_new mydb.dump
```

Directory, parallel

Use several jobs.

```
pg_restore -U postgres -j 4 \
  -d mydb_new mydb_dir
```

Handy flags

Clean, list, one table.

```
--clean --if-exists
pg_restore -l mydb.dump
pg_restore -t employee ...
```